

Math 300 Quiz 2 Spring 2024

Key

1. (3 pts) Use a complete sentence to give the definition of a rational number?

A rational number x has the form $x = \frac{a}{b}$ where a and b are integers and $b \neq 0$.

2. (3 pts) What does the zero product property say about real numbers?

The product of two nonzero real numbers is nonzero.

If x and y are real numbers with $x \cdot y = 0$, then either $x = 0$ or $y = 0$

3. (4 pts) Use a truth table to prove that $p \rightarrow q, \sim p \therefore \sim q$ is **not** a valid rule of deduction.

p	q	$p \rightarrow q$	$\sim p$	$\sim q$	$(p \rightarrow q) \wedge (\sim p)$	$[(p \rightarrow q) \wedge (\sim p)] \rightarrow (\sim q)$
T	T	T	F	F	F	T
T	F	F	F	T	F	T
F	T	T	T	F	T	F
F	F	T	T	T	T	T

This is not a valid rule of deduction since the last column is not a tautology.

In the third row, both premises are true but the conclusion is false.

Over $\rightarrow$

4. (5 pts) Prove for any integer n , if n^2 is even then n must be even.

Proof by contrapositive.

If n is odd then n^2 is odd.

Let $n = 2k+1$ for some k in $\mathbb{Z}$.

Then $n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ is

odd since $2k^2 + 2k$ is in $\mathbb{Z}$ whenever k is in $\mathbb{Z}$.

5. (5 pts) Prove that the argument $A \vee (\sim B), B, (A \wedge B) \rightarrow C \therefore C$ is valid. Give a reason for each step.

1. B premise
2. $A \vee (\sim B)$ premise
3. $\sim(\sim B)$ follows from 1 by double negation
4. A follows from 2 & 3 by disjunctive syllogism
5. $A \wedge B$ follows from 1 & 4 by conjunctive addition
6. $(A \wedge B) \rightarrow C$ Premise
7. C follows from 5 & 6 by modus ponens